Evolving a 2N3904 amplifier

A common-emitter stage evolved for 30 dB, 100 Hz–200 kHz and its bias point — CLAUDE.md §4, milestone M7

kronspice simulation report · 2026-11-07 · seed 1, population 100, 300 generations

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Summary

This is the milestone M7 example and the first run of the value-genome genetic algorithm (§4) on a nonlinear circuit. A common-emitter stage around a 2N3904 transistor (Fairchild datasheet Gummel–Poon model) is driven from a 600 Ω source into a 10 kΩ load on 12 V. Evolution chooses eight standard component values (resistors from E24, capacitors from E12) to meet a mid-band gain, both −3 dB band edges and a bias point.

- **Seed 1 meets the spec:** 29.92 dB at 10 kHz, band edges at 100.0 Hz and 199.8 kHz, V_{CE} 4.9 V, re-simulated from the written best.cir.
- **10 of 12 seeds pass** the acceptance test (30 ± 0.5 dB, edges within 5 %, V_{CE} ≥ 3 V). Seeds 3 and 10 miss the gain by less than 0.1 dB beyond the 0.5 dB band (29.44 and 29.41 dB).
- **The E-series grid sets a floor:** the weighted error settles near 0.02, and no single-part change of the best design improves it.
- **Energy ledgers balance:** the acceptance test requires the AC and OP ledgers to close to 10^{-9}.
- **Run time:** about 25 s for the full run; results are bit-identical for any worker count.

The circuit

circuit.cir is a plain SPICE deck: VCC 12 V; VIN with RS 600 Ω coupled through CIN into the base; the bias divider R1/R2; RC to the collector; RE1 in series with the bypassed RE2, so RE1 alone sets the mid-band gain (≈ RC || RL / RE1); CF from collector to base, the Miller capacitor that sets the upper edge; COUT 10 µF into RL 10 kΩ; and .ac dec 20 10 100meg.

The transistor is the datasheet's SPICE model (IS 6.734 fA, BF 416.4, VAF 74.03 V, CJC 3.638 pF, CJE 4.493 pF, TF 301 ps, RB 10 Ω). The datasheet print "Ise=6.734" is a known misprint for 6.734f and is corrected in the netlist. The engine solves the SPICE3 Gummel–Poon equations of device/bjt.go (forward and reverse Early effect, high injection, non-ideal base currents, depletion and diffusion charges, pnjlim limiting). The operating point is found by Newton with the M4 fallback chain, and the AC response is taken at a polished operating point (two extra Newton iterations so the small-signal transconductances are exact to round-off).

Genes, objectives and GA settings

gene	range	series	starting value (circuit.cir)
R1	10 kΩ – 1 MΩ	E24	47 kΩ
R2	1 kΩ – 100 kΩ	E24	10 kΩ
RC	470 Ω – 47 kΩ	E24	4.7 kΩ
RE1	10 Ω – 2.2 kΩ	E24	100 Ω
RE2	100 Ω – 10 kΩ	E24	1 kΩ
CF	1 pF – 1 nF	E12	10 pF
CIN	100 nF – 100 µF	E12	1 µF
CE	1 µF – 1 mF	E12	100 µF

objective	target	weight
gain at 10 kHz, V(o)/V(in)	30 dB	1
upper −3 dB edge	200 kHz	1

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objective	target	weight
lower −3 dB edge	100 Hz	1
collector voltage	6 V	0.2
emitter voltage	1.5 V (so V_{CE} ≈ 4.5 V)	0.2

GA settings: population 100, 300 generations, tournament 5, mutation rate 0.25 with σ = 0.1 decade, seed 1, 2 s timeout per evaluation, stop at a weighted error of 0.02. Operators (as implemented in M5): tournament selection, BLX-0.5 crossover on log magnitudes (Eshelman & Schaffer 1993), Gaussian mutation in decades, snapping to the E-series, elitism.

Results

Seed 1 after 300 generations

part	evolved value
R1 / R2	30 kΩ / 6.8 kΩ
RC	5.1 kΩ
RE1 / RE2	68 Ω / 1.3 kΩ
CF	33 pF
CIN / CE	4.7 µF / 18 µF

Re-simulated from the written best.cir: 29.92 dB at 10 kHz, edges at 100.0 Hz and 199.8 kHz, V_{CE} 4.9 V. best.cir opens in ngspice, LTspice and QSpice as it is.

Seeds 1–12 (doc/M7.md)

The acceptance test (evo/amplifier_test.go) holds 30 ± 0.5 dB, edges within 5 % and V_{CE} ≥ 3 V, the tolerance of the 5 % parts the design is made of.

seed	gain	lower edge	upper edge	V_{CE}	spec
1	**29.92 dB**	**100.0 Hz**	**199.8 kHz**	**4.91 V**	**met**
2	30.37 dB	100.1 Hz	201.3 kHz	4.62 V	met
3	29.44 dB	100.0 Hz	197.9 kHz	5.04 V	missed (gain)
4	30.18 dB	100.5 Hz	200.4 kHz	4.40 V	met
5	29.92 dB	100.0 Hz	199.9 kHz	4.35 V	met
6	29.96 dB	99.9 Hz	199.3 kHz	3.97 V	met
7	30.24 dB	100.6 Hz	199.6 kHz	4.97 V	met
8	29.90 dB	100.0 Hz	199.3 kHz	4.25 V	met
9	30.24 dB	99.9 Hz	200.0 kHz	4.27 V	met
10	29.41 dB	101.1 Hz	199.7 kHz	4.58 V	missed (gain)
11	29.80 dB	100.2 Hz	197.7 kHz	4.45 V	met
12	29.90 dB	99.6 Hz	197.6 kHz	4.63 V	met

How the search behaves

- **The E-series floor.** With E24/E12 values in about 4 % steps and five coupled targets, the weighted error settles near 0.02. No single-part change of the best design improves it, so that is the floor of the grid rather than of the search. A local search that moves several parts at once (the bias resistors together) would likely close the remaining gap; it is noted under M8 as future work.
- **Penalties, never a crash.** A candidate that fails to converge, gives a singular matrix, fails validation, breaks the ledger or times out gets a penalty fitness from the M5 classes (no-measurement < no-convergence < singular < invalid < ledger < timeout < other < panic), so the run keeps going.
- **Determinism.** One PCG stream per generation derived from the seed, and results stored by individual index, so a run is bit-identical for any worker count and resumes exactly from its checkpoint (§1 goal 5).

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Validation

- TestAmplifierExample (evo/amplifier_test.go) is the M7 acceptance test: the spec above plus AC and OP energy ledgers balanced to 10^{-9}.
- The export sheet for the starting circuit is MNA only: Kron's torn solution is shown for linear circuits, and the torn Newton solve (kron.JacobianTearing) is checked in kron's own tests.

Limits

- The design is a single stage evaluated at 27 °C; the temperature coefficients of the model are unused.
- Results are for 5 % E-series parts; the spec band (±0.5 dB, 5 % edges) is set to that tolerance.
- Two of twelve seeds miss the gain target by under 0.1 dB beyond the band; a multi-part local search is future work.

Reproducing the results

```
go run ./cmd/evolve -config examples/amplifier_gain/config.json   # about 25 s
go test ./evo -run TestAmplifierExample                            # the M7 acceptance test
python examples/amplifier_gain/viz/make_report.py examples/amplifier_gain/amplifier_gain_report.pdf
```

Outputs go to runs/<timestamp>-amplifier_gain/: a copy of the config, stats.csv (best, mean and worst fitness per generation), best.cir per generation and a final checkpoint.json that -resume continues from.

Sources: Fairchild 2N3904 datasheet (1997); Gummel & Poon, Bell Syst. Tech. J. 49, 827 (1970); Nagel, ERL Memo M520 (1975), ch. 4; Massobrio & Antognetti, Semiconductor Device Modeling with SPICE, 2nd ed., ch. 2; SPICE3f3 User's Manual §4.3; Eshelman & Schaffer, FOGA 2 (1993).

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